

Syllabus for CS 4200/ 6200 – Fall 2026 v1.24Jul2026

Intro to Quantum Computing, 3 credits

Class Time/Location: Asynchronous (No specific class meeting time)

Class Sections: CS4200 (section 88649) and CS6200 (section 88650) – taught in one lecture series – CS6200 has some extra material in addition to CS4200 materials

Course Description:

The objective of this class is to provide an “*introduction to quantum computing*.” We will cover terminology, brief introduction to quantum mechanics, linear algebra, classical algorithms, understanding of what is a qubit, 1-qubit operations, and 2-qubit gates, no-cloning theorem, 3-qubit gates, quantum algorithms, the potential of quantum computing to NP and NP-hard problems, code breaking, and optimization.

The overall goals of the course are the student should be able to know general terminology, understand basic concepts of quantum computing, know challenges to large-scale quantum computing, know who are the major contributors, and have an ability to read a current journal article in the field.

Form of Instruction: This is an online lecture course with textbook and journal article reading assignments, weekly homework and/or online quizzes. There are also two written midterm exams, and one final exam.

Course Goals and Learning Outcomes:

To understand fundamental techniques in designing quantum algorithms and knowing how to analyze them. This course will cover a broad range of quantum algorithms to include:

What is Quantum Computing (QC)?	Is QC real? and why its time is now!
Essential Quantum Physics (enough for QC)	Essential Linear Algebra (enough for QC)
Quantum Logic	Quantum Gates
Building Quantum Circuits	Introduction to Quantum Algorithms
Deutsch-Jozsa Algorithm	Code-Breaking
Simon's Algorithm	Grover's Algorithm
Shor's Algorithm	Quantum Annealing Machines

Assessment of these outcomes will be by written examinations and homework / quizzes.

Instruction and Class Attendance: This is ONLINE course. We do not meet in a classroom. You watch the lectures and videos online via PILOT, and have ability to interact with the instructor, Graduate TAs and other students via WebEx, text, email, or chat.

Office Hours: The specific office hour times are listed in the CONTENT section in a file marked Office Hours. Here are the Help Desk People.

- **Course Professor:** Nella Ludlow, PhD **Office:** Joshi 385 (when in Dayton)
Contact Info: Email: nella.ludlow@wright.edu **Personal Cell:** 360.202.8757
- **GTA: PhD Student – Vishnu Baligodugula** Email: baligodugula.2@wright.edu

You may also contact our Graduate TA about the course material and assignments. The Graduate TA will assist with questions and grading of assignments. If there is an error or omission, or some concern about lecture material, please contact me as I am the professor for the course. Please be kind and respectful to the GTAs.

Prerequisite: CS 3100 Data Structures, and Algorithms and MTH 2530 Linear Algebra

Some students may not have MTH2530--I do teach/review enough Linear Algebra in this course for you to understand the material. Please note that Quantum Computing and Quantum Algorithms do heavily rely upon Linear Algebra, and these concepts are needed to best learn this material.

I am also a Data Analytics and Mathematics Professor at Washington State University. So, if you have any questions or concerns about learning mathematics, please contact me. I can help you bridge math barriers. Please don't let a fear of math cause you to drop this course. I am confident we can find ways to help you learn this material!

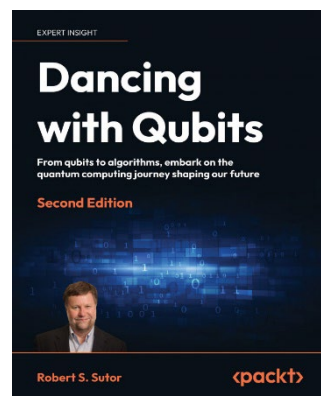
Textbook:

Dancing with Qubits 2nd Edition (1st Edition is good, but 2nd Edition is better)

(From qubits to algorithms, embark on the quantum computing journey shaping our future)

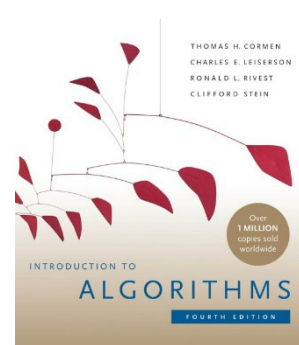
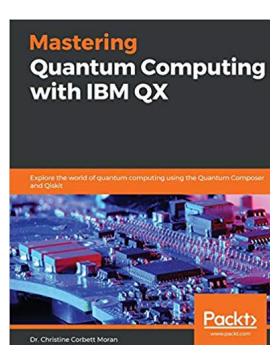
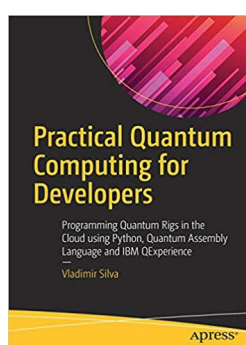
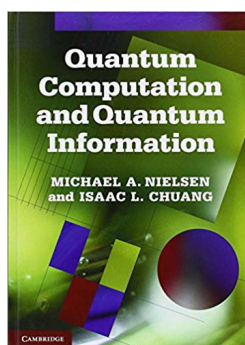
by Robert S. Sutor, 2024, ISBN **978-1837636754**

- Buy anywhere. Amazon Kindle prices is \$40.59
- <https://www.amazon.com/Dancing-Qubits-quantum-computing-change-dp-1837636753/dp/1837636753>
- <https://www.packtpub.com/en-us/product/dancing-with-qubits-9781837636754>



Optional Textbooks: This field is so new and progressing so rapidly that I will supply PDF copies of short sections from various textbooks, journal articles, and news media sources and post them in the Pilot blackboard.

Optionally, here are four good reference books:



- **Quantum Computation and Quantum Information, Nielsen and Chuang, 3rd Ed - 2017., Cambridge Press (ISBN-13: 978-1107002173)**
- **Practical Quantum Computing for Developers, Vladimir Silva, 1stEd - 2018., Apress (ISBN-13: 978-1-4842-4217-9-53799)**
- **Mastering Quantum Computing with IBM QX, Christine Moran, 1stEd - 2019., Packt (ISBN-13: 978-1-78913-643-2)**
- **Introduction to Algorithms, Cormen, Leiserson, Rivest, and Stein, 4th Ed., MIT Press (ISBN: 9780262046305)**

Final Exam: Anytime during Monday, December 7th, 2026
Two hours in length. Final is cumulative. Exam is online.

Assignments and Exams: There will be two online midterm exams and the scheduled final exam which will be cumulative. Homework or online quizzes be assigned and submitted through Pilot approximately bi-weekly (approximately once every two weeks).

Total points will be determined by a weighted average of the student's scores on Homework/Quiz, Project, Midterm Exams, and the Final Exam as shown below.

• Quizzes	20%
• Quantum Program Assignments	20%
• Midterms (two at 15% each)	30%
• Final Exam	30%

The CS4200 and CS6200 sections are taught together. There are some areas where CS6200 has extra reading and an extra assignment. See CS6200 Requirement below.

CS6200 Additional Requirement: CS6200 Graduate Students have some extra work, which are to review a Quantum Computing journal article assigned by the instructor and answer questions requiring deeper understanding on a graduate version of Midterm 2. You will also have a quantum gates homework programming assignment.

Extra Credit: We may provide an extra credit opportunity. Such as attend a special guest lecture, or read a short book written by Nobel Laureate in Quantum Physics, or to watch a movie by a world's great algorithm designer, et cetera. This allows us to add outside material, give second opportunities to struggling students, to challenge the super star students. Extra credit is optional, and only provides a few additional points to HW/Quiz score.

Grading Policy: Course grades are based on a weighted point system. The grading scale is a standard A-B-C-D-F based upon a weighted percentage score:

A : 90.00 – 100.00
B : 80.00 – 89.99
C : 70.00 – 79.99
D : 60.00 – 69.99
F : 0 – 59.99

However, at the discretion of the Instructor, the grading scale may be “relaxed” if deemed appropriate, but will be applied to all students equally!

Late Work Policy: Late submissions are not normally allowed. However, earlier submissions are allowed at any time before due date. Extensions *may* be allowed by contacting the instructor well in advance. After “answer keys” have been posted to the class, that assignment or exam cannot be made up, so communicate to the instructor and don't get too far behind. If asking an extension, do so on real emergencies—not as a habit.

It is better to turn in a partial homework than no homework. The phrase “*little is little but nothing is nothing*” is applicable. For an excused missed midterm exam, the accommodation is at the sole discretion of the instructor and may include a reweighting of the remaining components making up the student's grade or taking a makeup exam.

WSU Classroom Safety: Although this class is online, safety is still important. Please review Wright State University safety procedures at <https://www.wright.edu/public-safety/emergency-preparedness>.

I highly recommend you sign up for the emergency text message service. Please [visit the Wright State University's Alert website](#).

Students with Disabilities: Accessible Information Management (AIM) is a web-based tool for Wright State students registered with the Office of Disability Services(ODS). AIM currently provides students the following services:

- request ODS services each semester
- electronically e-mail accommodation letters to your professors
- schedule any exams or quizzes that you want to complete in the ODS Test Proctoring Center

More information may be found here: <https://www.wright.edu/diversity-and-inclusion/disability-services/aim-student-access>

Academic Integrity Policy: All homework must be done individually. Anyone cheating will receive a zero for that assignment *and* will be subject to the university's academic dishonesty/integrity policy, which may include being expelled from the course and receiving an F. For more information read this link <https://policy.wright.edu/policy/3710-academic-integrity-standards-and-process-misconduct>

Use of Generative AI Policy: You are not permitted to use generative AI tools for any work for this section of this course. This includes the use of popular tools like *ChatGPT*, *Codex*, *Claude*, *Midjourney*, *GitHub Co-Pilot*, as well as all other tools built on generative AI technologies. Due to the nature of this course, the professor can only fairly and accurately evaluate work that is not assisted by generative AI. Use of generative AI for assigned work in this course will be considered a violation of the university's academic integrity policies. If you have any questions about whether or not use of a particular tool or technology is allowed, check with your instructor first.

Disclaimer: Although unlikely, please note that the policies established in this syllabus could potentially change. Any changes will be formally announced in class and on the Wright State Pilot Blackboard for this course. It is the student's responsibility to attend class (watch lectures) and to check regularly course website (pilot.wright.edu).

Flexibility: Please be flexible. We *may* change the schedule slightly, allowing more or less time on an individual topic based on the class as a whole's understanding. In every class, we will have some students wanting to move faster, and some needing more time. As instructor, I reserve the right to make small adjustments in the best interest of the class.

My top goal is to help You learn: This course is about an extremely exciting field. I am very passionate about this topic and honored to introduce YOU to Quantum Computing!

Course Outline

This is an approximate schedule of the course. The timing may vary slightly.

Week	Topic
Week "0"	Review Syllabus, Class Structure, & Instructor Background
Week 1	What is Quantum Computing (QC)
Week 2	Why it is real and Why its time is now (Professor to submit Participation Roster by September 1 st , 2026)
Week 3	Quantum Physics and Linear Algebra for QC (Labor Day Holiday – September 7 th , 2026)
Week 4	Quantum Logic
Week 5	Quantum Gates
Week 6	Review for Midterm 1 (Midterm 1 – Thursday, October 1 st , 2026)
Week 7	Quantum Computing and Prohibited Maneuvers (Professor to submit Mid-Semester Grades by October 12 th , 2026)
Week 8	Building Quantum Circuits
Week 9	Introduction to Quantum Algorithms
Week 10	Deutsch and Deutsch-Jozsa Algorithms
Week 11	Simon's Algorithm and Grover's Algorithm
Week 12	Review for Midterm 2 (Midterm 2 – Thursday, November 12 th , 2026)
Week 13	Shor's Algorithm and Code-Breaking
Week 14	More on Shor's Algorithms (Thanksgiving Break – No Classes November 25 thru 27, 2026)
Week 15	Course Review
Finals	Final Exam – Monday, December 7 th , 2026 (Professor to submit Final Grades by Wednesday, December 16 th , 2026)